

15.88M

Total Revenue

99K

Total Orders

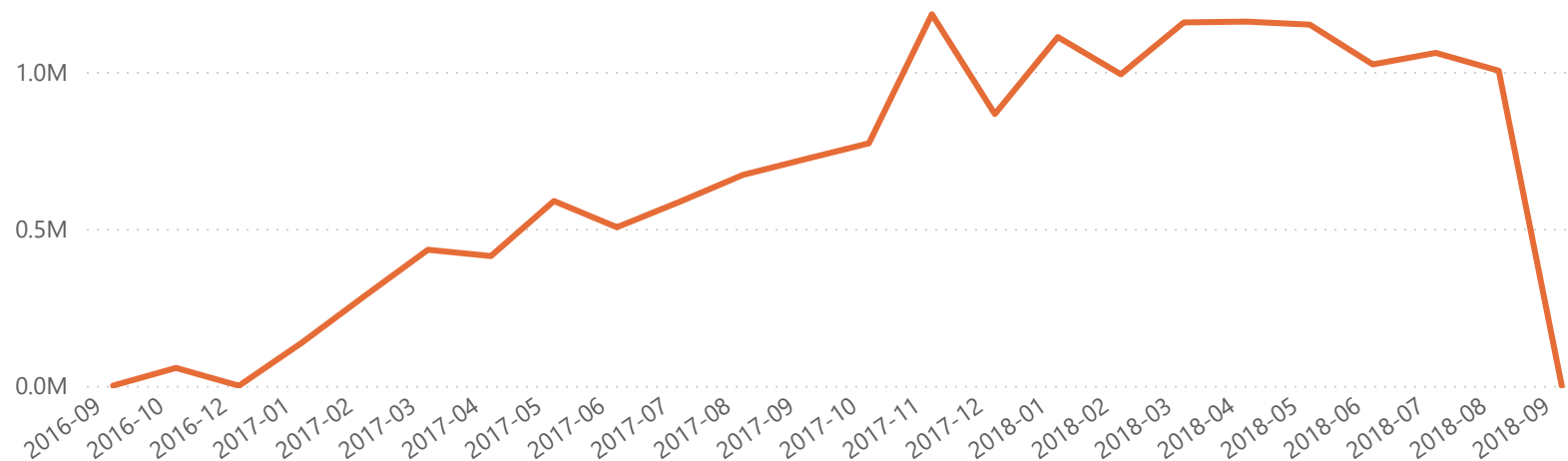
95K

Unique Customers

4.03

Average Review Score

Total Revenue by Year Month



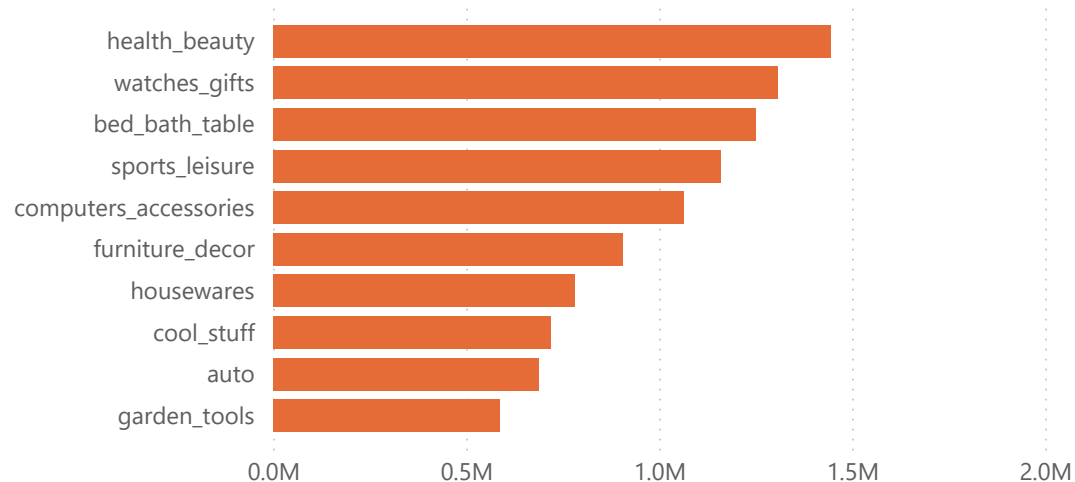
Executive Summary

This project analyzes the Olist Brazilian e-commerce marketplace dataset using Python, SQL Server, and Power BI. The workflow includes data cleaning, SQL-based analytical modeling, and dashboard development.

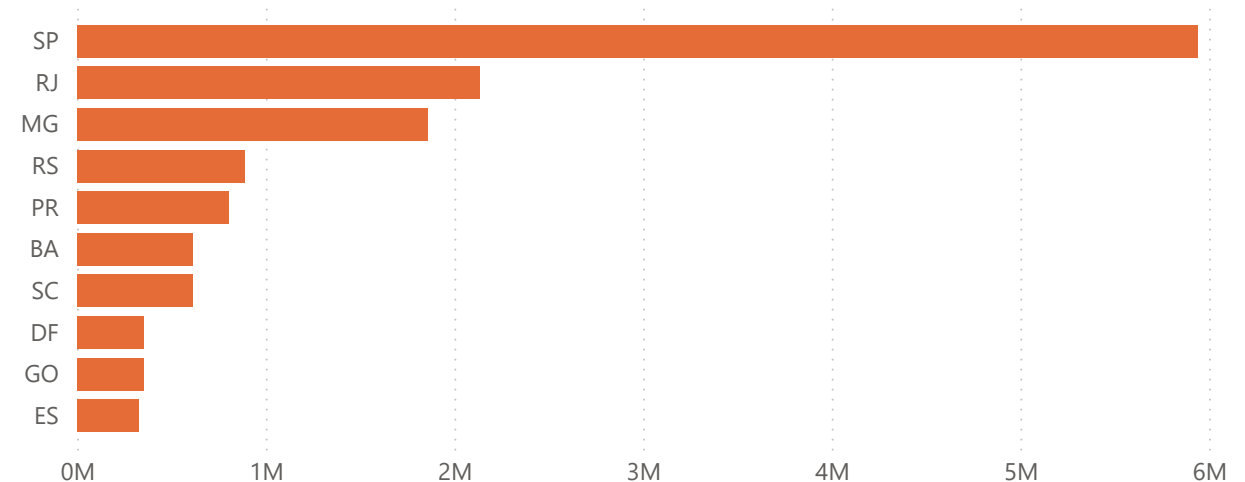
Key findings:

- Revenue and order activity vary significantly across product categories and customer states.
- Late delivery is associated with lower customer review scores and higher low-review rates.
- Several states show both elevated late-delivery rates and weaker customer satisfaction, suggesting regional logistics bottlenecks.
- Customer repeat purchasing is weak with

Total Revenue by Category



Total Revenue by State



Seller
Performa...

Cohort
Analysis

Review
Impact

Delivery
Performa...

Executive

Delivery Performance & Customer Satisfaction

Analyzing how delivery delays vary by region and how they relate to review outcomes.

95K

Delivery Orders

7.99%

Late Delivery Rate

12.0

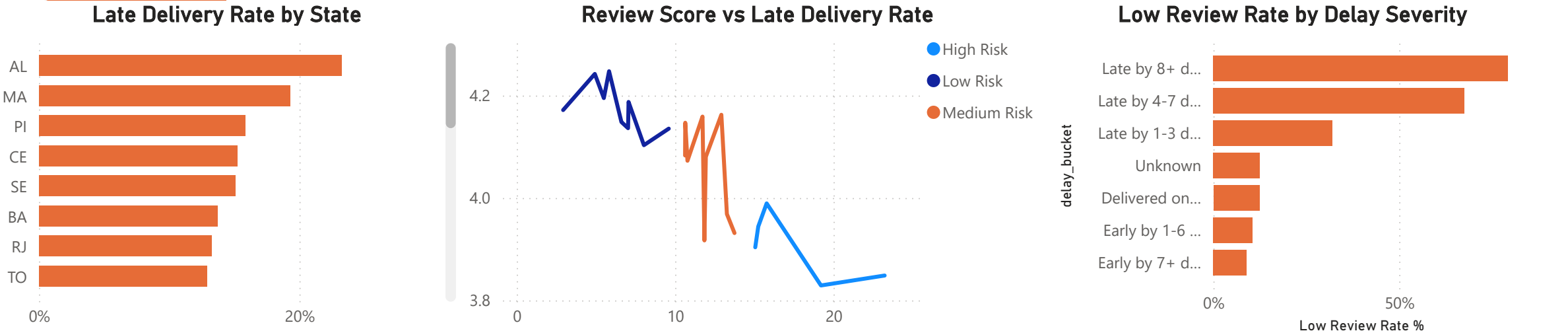
Average Delivery Days

-11.9

Average Delay Days

4.03

Average Review Score



Year Month

customer_state

delivery_status

All

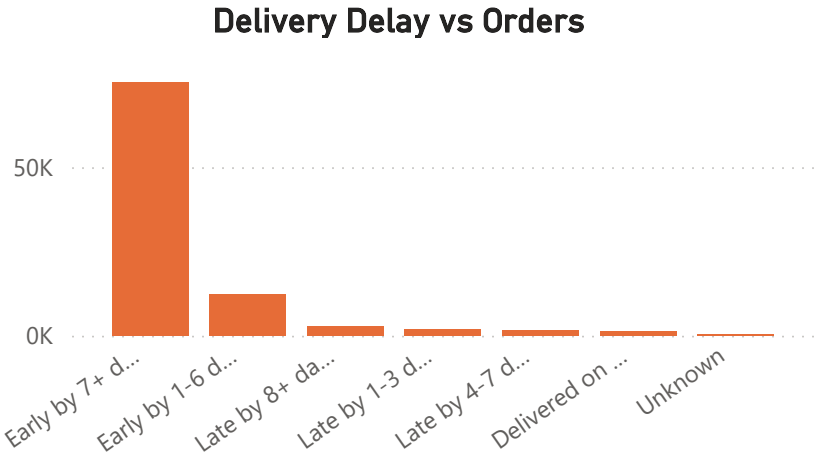
All

Late

On Time or Early

Key Insight: Late deliveries are associated with lower customer review scores and higher low-review rates. Several states show both elevated late-delivery rates and weaker customer satisfaction, suggesting regional

Regional Logistics Risk								
State	Total Orders	Late Delivery Rate	Avg Delivery Days	Avg Delay Days	Avg Review Score	Low Review Rate	Logistics Risk	
BA	3213	13.75	18.78	-10.88	3.93	17.01	Medium Risk	
ES	1963	11.94	15.16	-10.68	4.08	13.77	Medium Risk	
MS	695	11.73	15.21	-11.00	4.16	13.30	Medium Risk	
PA	924	11.85	23.20	-14.13	3.92	17.89	Medium Risk	
PB	510	10.78	19.78	-13.45	4.07	14.90	Medium Risk	
PE	1576	10.63	17.88	-13.38	4.08	14.81	Medium Risk	
RJ	12141	13.27	14.78	-11.83	3.97	18.23	Medium Risk	
Total	24932	196.04	266.44	-161.88	56.04	228.29		



Review Impact Analysis

Key Insight: Review scores decline as delivery delays become more severe. Late orders show a higher share of low reviews indicating that delivery reliability is closely associated with customer satisfaction.

95K

Reviewed Orders

4.16

Avg Review Score

12.78%

Low Review Rate

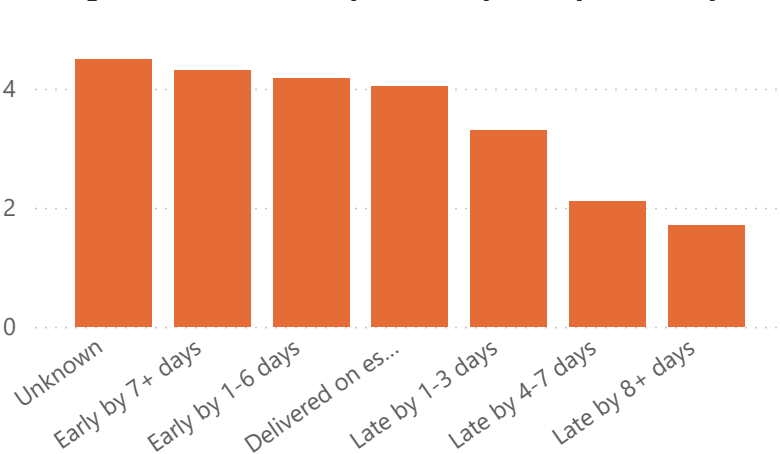
79.04%

High Review Rate

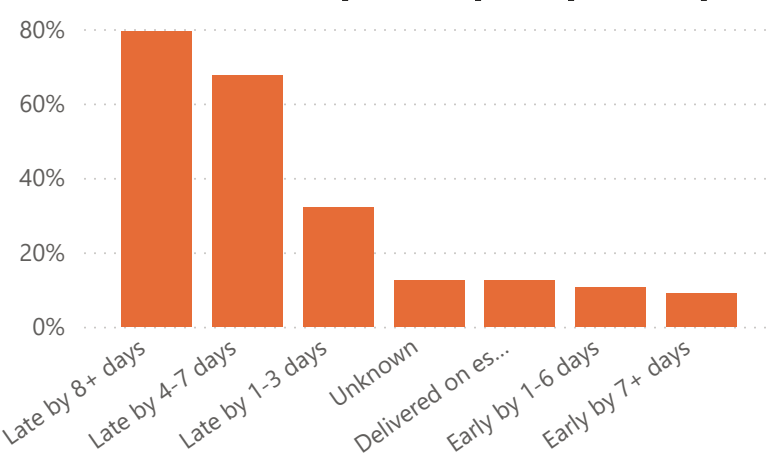
9.73%

One-Star Rate

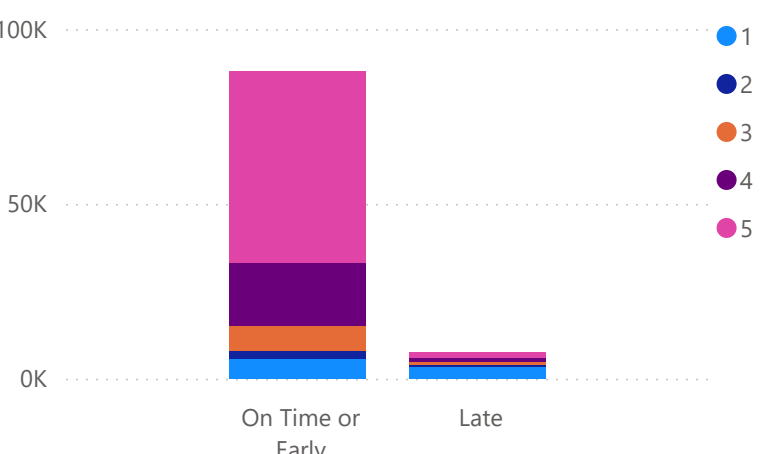
Avg. Review Score by Delivery Delay Severity



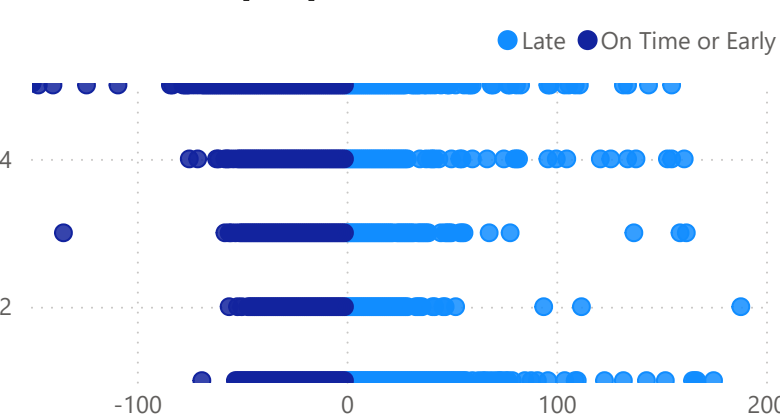
Low Review Rate by Delivery Delay Severity



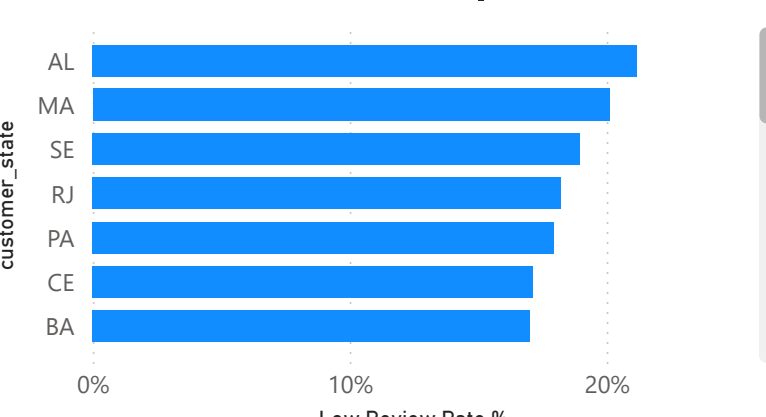
Review Score Distribution: Late vs On-Time Orders



Delay Days vs Review Score



Low Review Rate by State



Delay Bucket	Delivery Avg Review Score	Low Review Rate
Delivered on estimated date	4.04	12.41%
Early by 1-6 days	4.18	10.58%
Early by 7+ days	4.31	8.97%
Late by 1-3 days	3.30	31.99%
Late by 4-7 days	2.11	67.57%
Late by 8+ days	1.70	79.33%
Unknown	4.50	12.50%
Total	4.16	12.78%

Year Month

All

customer_state

All

delivery_status

Late

On Time or Early

Customer Retention Overview

93K

Total Customers Cohort

96K

Customer Orders

3K

Repeat Customers

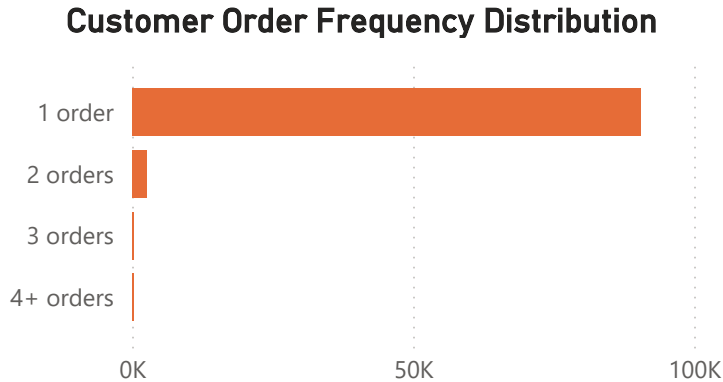
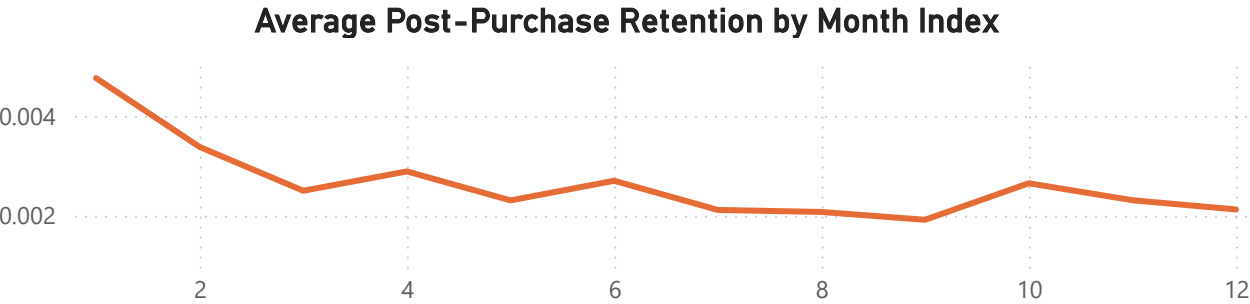
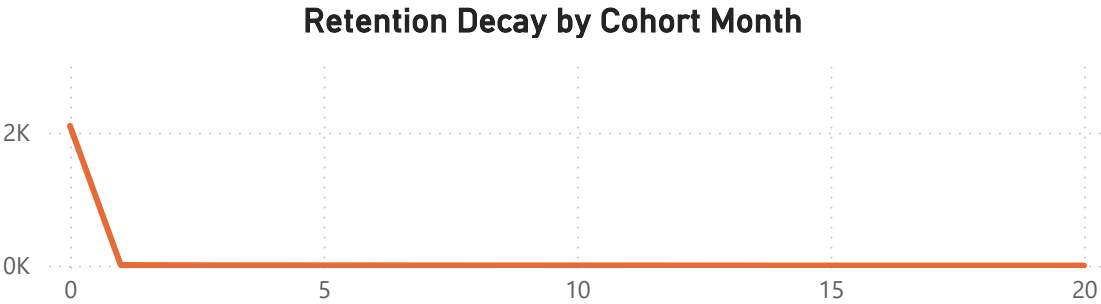
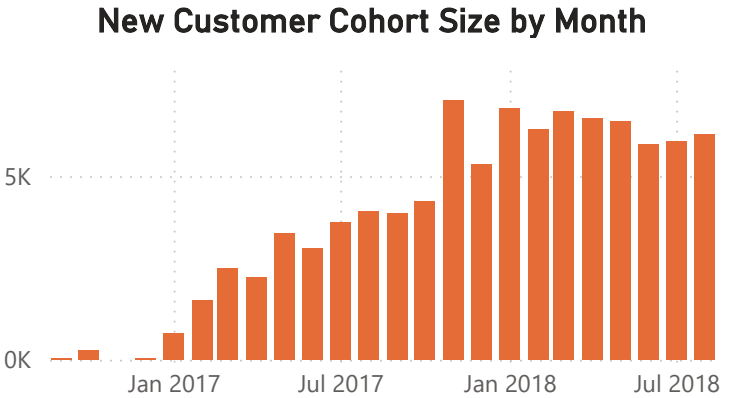
3.00%

Repeat Customer Rate %

1.03

Avg Orders per Customer

cohort_month	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
01 September 2016	100.00																	
01 October 2016	100.00						0.38			0.38		0.38		0.38		0.38		0.38
01 December 2016	100.00	100.00																
01 January 2017	100.00	0.28	0.28	0.14	0.42	0.14	0.42	0.14	0.14		0.42	0.14	0.70	0.42	0.14	0.14	0.28	0.42
01 February 2017	100.00	0.18	0.31	0.12	0.43	0.12	0.25	0.18	0.12	0.18	0.12	0.31	0.12	0.18	0.12	0.06	0.06	0.18
01 March 2017	100.00	0.44	0.36	0.40	0.36	0.16	0.16	0.32	0.32	0.08	0.36	0.12	0.20	0.12	0.16	0.24	0.08	0.12
01 April 2017	100.00	0.62	0.22	0.18	0.27	0.27	0.35	0.31	0.31	0.18	0.27	0.09	0.04	0.04	0.09	0.09	0.13	
01 May 2017	100.00	0.46	0.46	0.29	0.29	0.32	0.41	0.14	0.26	0.26	0.26	0.35	0.23	0.03	0.17	0.20		



cohort_month

01-09-2016

01-08-2018

cohort_index

0

20

Key Insight: Customer repeat purchasing is weak in the Olist marketplace. Only about 3% of customers placed more than one delivered order. Cohort retention after the acquisition month remains low, and small cohorts should be interpreted carefully because they can inflate retention percentages.

Seller Performance Overview

3K

Total Sellers

15.45M

Seller Revenue

93

High Risk Sellers

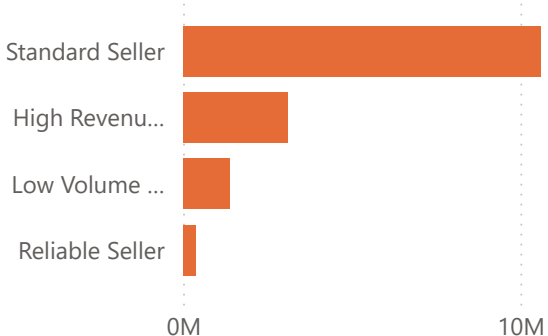
26

Reliable Sellers

4.15

Avg Seller Review Score

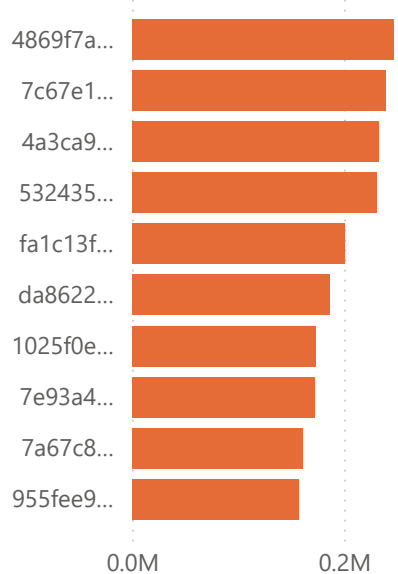
Revenue by Seller Risk Segment



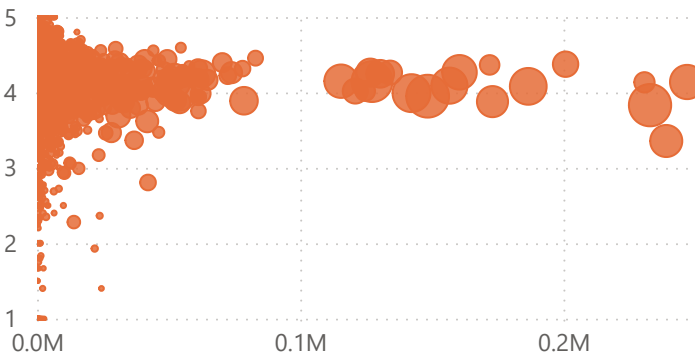
Seller Count by Segment



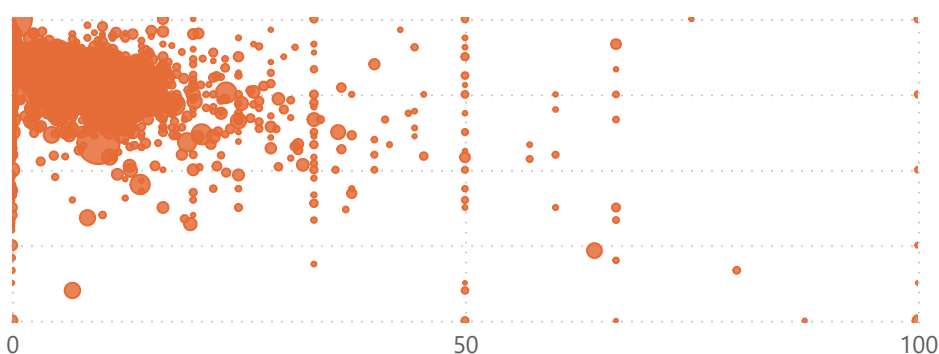
Top 10 Sellers by Revenue



Seller Revenue vs Review Score



Seller Delivery Risk vs Review Score



seller_segment

- High Revenue / High Risk
- Low Volume Seller
- Reliable Seller

Key Insight: Seller performance is uneven across the marketplace. A subset of sellers contributes meaningful revenue while also showing weaker review scores, higher late-delivery rates, or higher low-review rates. These sellers represent marketplace quality risk and should be prioritized for operational review.